LevelUP

AI-Powered Career Operating System

AI-Powered Engineering Platform	Multimodal Voice + Vision + GitHub Inspector
MiniMax-M3 LLM · FastAPI · React 18	1-Click Kanban Auto-Schedule · Human-in-the-Loop Safety

GitHub (Primary)	github.com/Prajeeth-12/LevelUP-new
GitHub (Secondary)	github.com/Prajeeth-12/LevelUP_HCL
Live App	level-up-new.vercel.app
Branch	new-theme-layrs
Submission Date	August 29, 2026
Tech Stack	FastAPI · React 18 · Firebase · MiniMax-M3

1. Executive Summary

LevelUP is an **AI-powered personal career operating system** designed to bridge the modern engineering skill gap. It transforms passive learning into an active, AI-guided experience with autonomous skill gap analysis, voice-first tutoring, multimodal vision inspection, and 1-click Kanban sprint scheduling.

Core Problems Solved

- **Skill Gap Blindness**: Developers lack tools to objectively benchmark their current skills vs. target job requirements.
- **Passive Learning**: Traditional courses don't adapt to individual cognitive bandwidth or weekly availability.
- **Task Overload**: Unstructured backlog leads to burnout and lost learning momentum.
- **Isolation**: Most platforms offer no real-time AI mentor for on-demand technical coaching.

LevelUP's Answer: A proactive AI agent that runs autonomous skill analysis, speaks aloud with the user, organizes their Kanban board into daily sprints, inspects GitHub repositories for architectural gaps, and always asks for human approval before saving changes — ensuring the learner stays in full control.

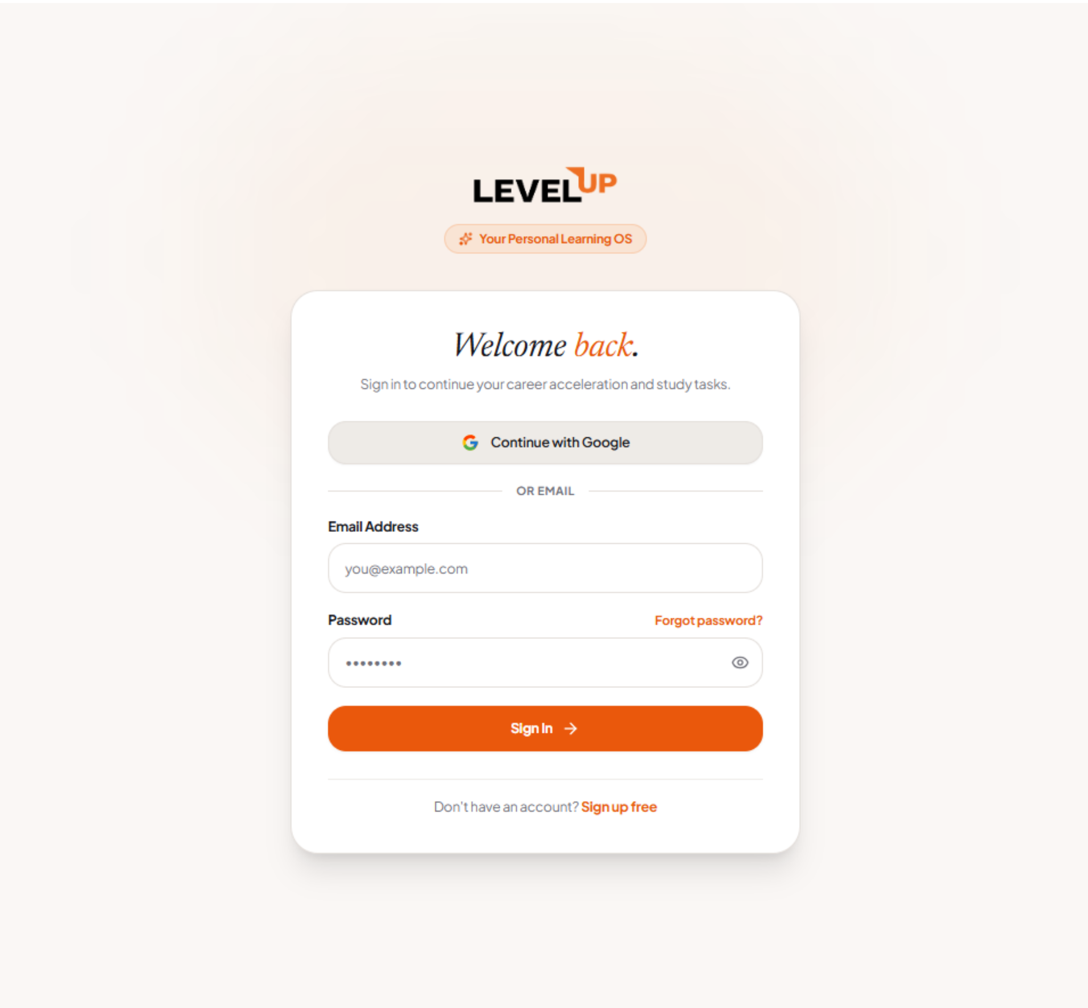

Figure 1: LevelUP Sign In page — Layrs warm minimalist design with editorial serif typography

2. System Architecture

LevelUP is a modular, full-stack platform with four primary layers: a React SPA frontend, a FastAPI async backend, the MiniMax-M3 LLM AI engine, and Google Cloud Firestore as the real-time database. All components communicate via authenticated REST and WebSocket protocols.

```
[Browser: React 18 + Vite]
↕ HTTPS REST
[Backend: FastAPI + Uvicorn]
↕ OpenAI-Compatible API
[AI Engine: MiniMax-M3 LLM @ api.gmi-serving.com]
↕ Firebase Admin SDK
[Database: Google Cloud Firestore]
[Auth: Firebase Authentication]
```

Figure 2: High-level system architecture diagram

Technology Stack

Layer	Technology	Role
Frontend	React 18 + Vite + Tailwind CSS	SPA, routing, Kanban UI, responsive design
Styling	Layrs Design System	Custom warm palette, editorial typography
Backend	FastAPI + Uvicorn (Python 3.10)	REST API, async routes, prompt orchestration
AI Engine	MiniMax-M3 LLM (GMI Serving)	High-context reasoning, JSON action extraction
Voice	Web Speech API (STT + TTS)	Bidirectional voice tutoring in-browser
Database	Google Cloud Firestore	Real-time NoSQL skill/task/roadmap storage
Auth	Firebase Authentication	Google OAuth + email/password + RBAC
Hosting	Vercel + uvicorn	Frontend CDN + Python backend server

Data Flow: AI Chat Request

Step	Action	Layer
1	User types or speaks a message	Frontend (React)
2	STT converts audio → text (Web Speech API)	Browser API
3	POST /ai/chat with user message + Firebase JWT	FastAPI Backend
4	Backend verifies token, loads user context from Firestore	Backend + Firestore
5	System prompt + context injected into MiniMax-M3 call	Backend → LLM API
6	LLM returns rich Markdown + JSON action cards	MiniMax-M3 LLM
7	Backend parses actions (CREATE_SKILL, MOVE_TASK, etc.)	Backend

8	Response streamed back as JSON to frontend	FastAPI → React
9	ChatMessageRenderer renders Markdown + Approval Cards	Frontend (React)
10	User clicks Approve → changes written to Firestore	Firestore

3. Authentication & Onboarding

LevelUP uses Firebase Authentication with Google OAuth and email/password support. The authentication UI was redesigned from scratch to match the Layrs warm minimalist design system — featuring editorial serif typography, warm cream canvas, and the signature orange primary color.

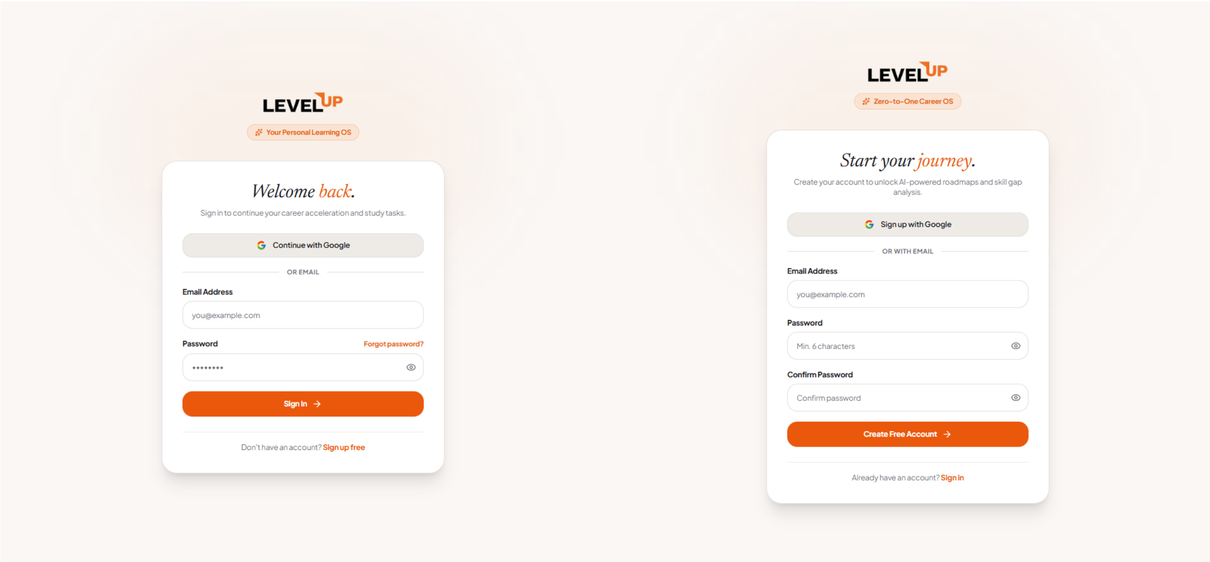

Figure 3: Left — Sign In page. Right — Sign Up (Create Account) page.

Design Decisions

- **Logo Adaptive Mode**: Uses /logo.png in light mode and /logo-white.png in dark mode.
- **Editorial Headline**: Serif italic "Welcome *back*." communicates warmth and familiarity.
- **Google OAuth First**: Primary CTA reduces friction; email form available as secondary option.
- **Password Reset Flow**: Inline reset mode within the same card — no page navigation needed.

3.5 Dashboard — App Overview

The LevelUP dashboard is the learner's home base with a daily question prompt, AI roadmap cards, cycle stats (In Progress, Today's Tasks, Mastered, Readiness Score), and full sidebar navigation.

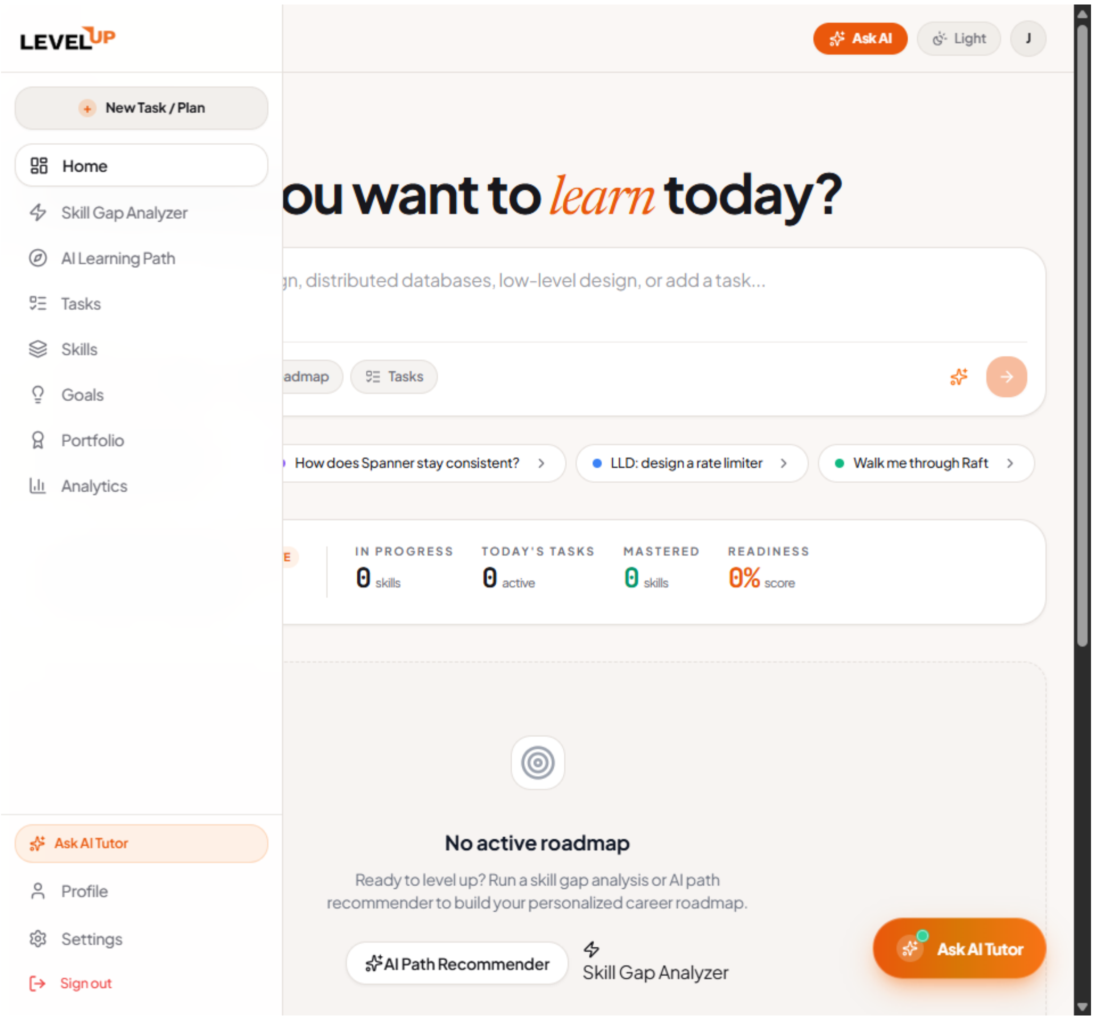

Figure 4: Dashboard — Home view with sidebar navigation, cycle stats, and AI roadmap panel

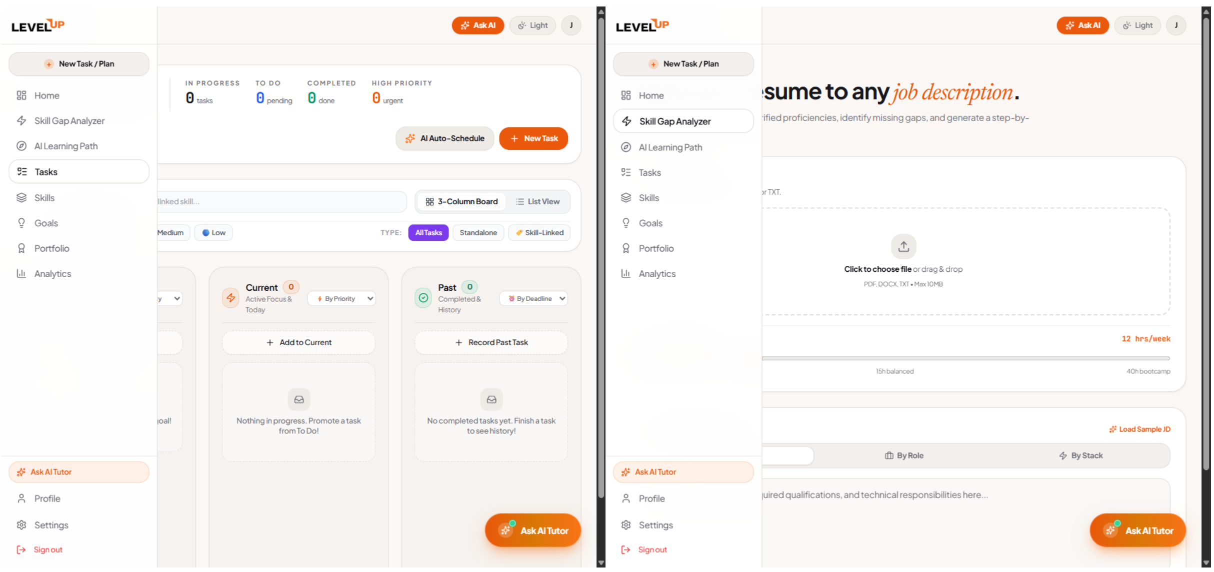

Figure 5: Left — Kanban Task Board with AI Auto-Schedule button. Right — Skill Gap Analyzer.

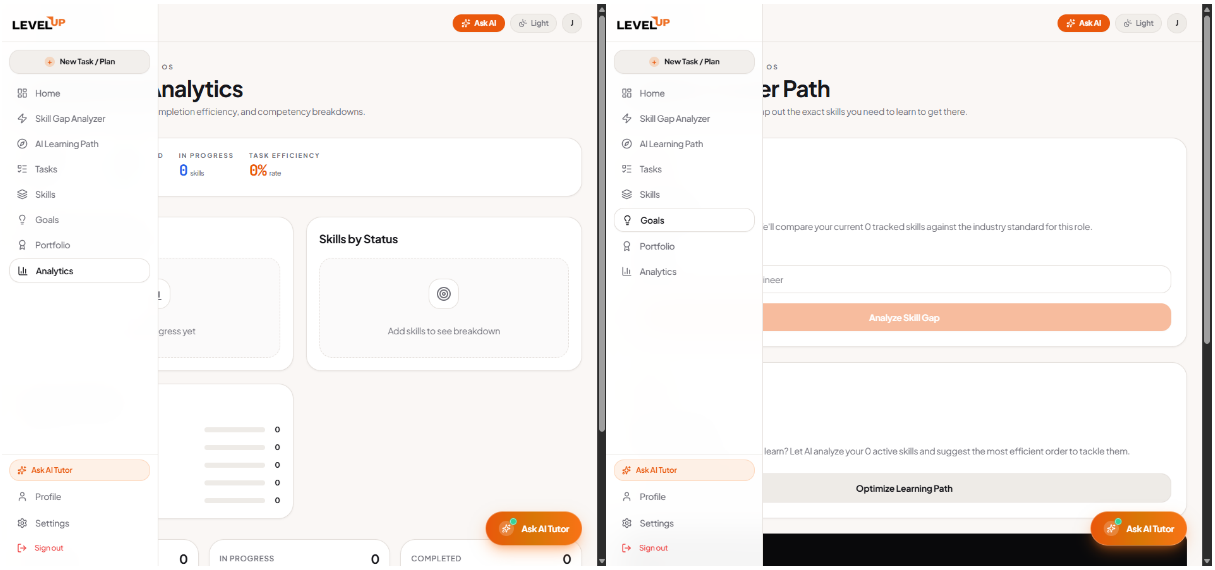

Figure 6: Left — Analytics page with progress charts. Right — Goals & Milestones tracker.

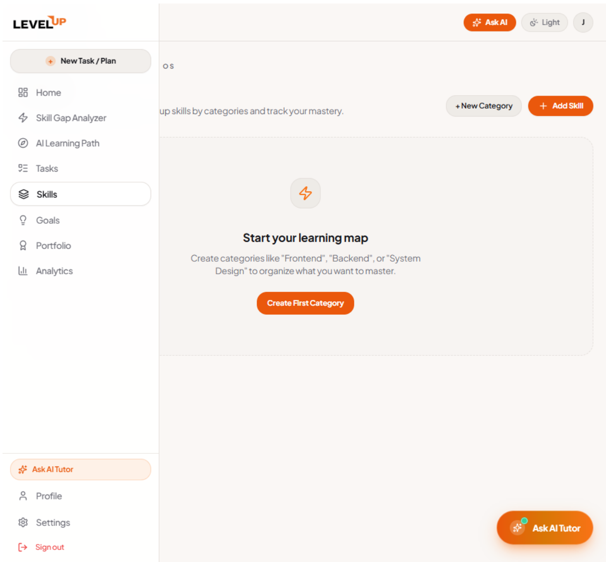

Figure 7: Skills board — categorized skill cards with progress indicators and priority badges

4. Core Application Features

4.1 AI Chat Assistant with MiniMax-M3 LLM

The AI Chat Drawer is the heart of LevelUP. Powered by MiniMax-M3 LLM via the OpenAI-compatible endpoint at api.gmi-serving.com, it provides real-time technical mentoring, architectural guidance, and proactive workspace management through structured JSON action cards.

Feature	Description
Live LLM Inference	MiniMax-M3 LLM calls via GMI Serving API with verified 200 OK responses
Rich Markdown Output	Code blocks with 1-click copy, bold formatting, numbered lists, tables
JSON Action Cards	AI proposes CREATE_SKILL, MOVE_TASK, ADD_ROADMAP as interactive approval cards
Context Awareness	Injects user's skills, tasks, and roadmap progress into every LLM system prompt
Fallback Intelligence	Rule-based parser handles edge cases when LLM is unavailable

4.2 Voice-First 1-on-1 AI Tutor

Bidirectional voice interaction built on the Web Speech API. Users can ask technical questions aloud and receive spoken turn-by-turn answers — ideal for hands-free studying during coding sessions.

Feature	Description
Speech Recognition	Live microphone input with animated waveform indicator in the chat drawer
Speech Synthesis	Natural-sounding TTS playback of every AI response when Voice Tutor is ON
Instant Stop Control	One-click stop button interrupts audio playback mid-sentence
Toggle Header Button	Volume icon in drawer header enables/disables voice mode globally

4.3 1-Click Kanban Smart Auto-Scheduler

The AI Auto-Schedule system analyzes all tasks across the 3-column Kanban board (To Do, Current, Past), calculates weekly cognitive bandwidth from a user-adjustable slider (4–40h/week), and proposes an optimized daily sprint with priority ordering.

Feature	Description
Bandwidth Calibration	Users set weekly study hours (4h–40h) via a slider before scheduling
Sprint Prioritization	AI ranks tasks by urgency, effort, and learning impact
Human Approval Grid	Interactive checkbox list — user confirms each proposed move before applying
Backend Route	POST /ai/auto-organize-tasks with task payload and bandwidth budget

4.4 Multimodal Vision & GitHub Architecture Inspector

Users can drag-and-drop architecture diagrams, whiteboard sketches, or database ERDs directly into the AI Chat. Alternatively, pasting a GitHub URL triggers automated repo metadata extraction and architectural gap analysis.

Feature	Description
Image Upload	Supports .png, .jpg, .webp architecture diagrams via drag-and-drop in chat
GitHub Scraping	Pasting a GitHub URL fetches repo name, description, language, and star count
Vision Analysis	AI identifies missing engineering patterns from diagrams or repo structure
Proposed Action Cards	Extracted skills and tasks rendered as interactive approval cards in chat

4.5 Skill Gap Analyzer

A 2-step intake process where users upload their resume and specify a target job description. The AI calculates match scores, identifies skill gaps, and proposes a prioritized roadmap. Results are displayed with match score cards, phase milestones, and next-action recommendations.

Feature	Description
Resume Upload	PDF/text resume parsing with AI-driven skill extraction
JD Matching	Three evaluation axes: Job Description, Tools & Technologies, Domain Concepts
Match Score Cards	Visual percentage cards for each evaluation axis with color coding
Phase Milestones	5-phase learning roadmap generated from gap analysis results
Cycle Stats Bar	Real-time metrics: days active, skills tracked, tasks completed

4.6 Human-in-the-Loop Safety System

Every AI-proposed workspace action — whether creating a skill, moving a task, or adding a roadmap — goes through a human approval gate. The AI generates interactive Proposed Workspace Action Cards that display the change, rationale, and an Approve & Save button. No data is written to Firestore until the user explicitly approves.

5. UI/UX Design System — Layrs Theme

LevelUP uses a custom design system called 'Layrs' — a warm, editorial visual language inspired by high-end publishing and premium productivity apps. Every component was custom-designed and built with Tailwind CSS utility classes mapped to CSS custom property tokens.

Token	Hex Value	Usage
Primary Orange	#EA580C	Primary CTA buttons, links, headings accent, active states
Obsidian Dark	#0B0C11	App background (dark mode), navigation, contrast elements
Cream Canvas	#FBF7F4	App background (light mode), cards, editorial warmth
Border Default	#E2D9CF	Card borders, dividers, form field outlines
Emerald	#10B981	Success states, completed tasks, positive metrics
Amber	#F59E0B	Warning states, in-progress tasks, attention callouts
Muted Foreground	#6B7280	Secondary text, captions, placeholders

Typography System

- **Display / Hero**: Serif italic (font-serif italic) for emotional headline moments — 'Welcome back.', 'Start your journey.'
- **Body / UI**: Inter / system-ui — clean, readable sans-serif for all interface text
- **Monospace**: JetBrains Mono / Courier — for all code blocks in the AI chat renderer

Component Design Patterns

- **Cards**: rounded-3xl corners, border-border outline, elevated with shadow-xl
- **Buttons**: rounded-2xl with micro-hover transitions (bg-orange-600 → bg-orange-700)
- **Inputs**: rounded-2xl with focus ring in orange, placeholder in muted-foreground
- **Tags & Chips**: pill-shaped with orange/emerald/amber semantic color coding
- **Navigation**: Sidebar rail with obsidian dark background, white icon labels, orange active indicator

6. Competition Submission Checklist

#	Deliverable	Location
1	Source Code (ZIP)	C:\Users\praje\Downloads\LevelUP_SourceCode.zip (1.37 MB)
2	GitHub Repo (Primary)	github.com/Prajeeth-12/LevelUP-new (branch: new-theme-layrs)
3	GitHub Repo (Secondary)	github.com/Prajeeth-12/LevelUP_HCL (branch: new-theme-layrs)
4	Solution PDF	C:\Users\praje\Downloads\LevelUP_Solution_Documentation.pdf
5	Demo Video URL	https://youtu.be/[TO-BE-RECORDED]
6	Deployed App URL	https://level-up-new.vercel.app

7. Local Setup & Execution Guide

Prerequisites

- Node.js v18+ installed → node -v
- Python 3.10+ installed → python --version
- Git installed → git --version

Backend Setup

```
cd backend
python -m venv venv
.\venv\Scripts\activate # (Windows)
pip install -r requirements.txt
# Configure backend/.env with MINIMAX_API_KEY, FIREBASE config
python -m uvicorn app.main:app --port 8000 --reload
```

Frontend Setup

```
cd frontend
npm install
npm run dev -- --port 3000
# Open browser at: http://localhost:3000
```

Verification Checklist

- Open http://localhost:8000/docs — FastAPI Swagger UI should load
- Open http://localhost:3000 — LevelUP app should load with Layrs theme
- Sign in with Google or email/password
- Open AI Chat Drawer and ask a technical question — MiniMax-M3 should respond
- Go to /tasks and click 'AI Auto-Schedule' — sprint proposal should appear
- Go to /skill-gap and enter a job description — gap analysis should run

8. Key API Endpoints

Method	Endpoint	Description
POST	/ai/chat	Main LLM chat endpoint with MiniMax-M3
POST	/ai/auto-organize-tasks	1-Click Kanban Auto-Schedule with bandwidth budget
POST	/ai/skill-gap-analysis	Resume vs. JD skill gap analysis
GET	/skills	List all user skill categories and items
POST	/skills	Create a new skill category
POST	/skills/{id}/items	Create a new skill item within a category
PUT	/skills/{id}/items/{iid}	Update skill progress, status, notes
GET	/tasks	List all Kanban tasks across all columns
POST	/tasks	Create a new Kanban task
PUT	/tasks/{id}	Update task status, priority, assignment
GET	/roadmap	Get user's AI-generated learning roadmap
POST	/roadmap/generate	Generate new roadmap from skill gap analysis

9. Git Repository & Commits

Key commits on the new-theme-layrs branch:

Commit Hash	Description
6f08efc7	feat: Complete theme harmonization across SignIn, SignUp, ProfileSetup, and all internal pages
667209bb	docs: Add LaTeX solution documentation, figures, and competition submission package
c5f9d9ba	fix: Enforce MiniMax-M3 LLM API key loading with dotenv override - verified direct LLM connectivity
9fc82333	feat: Add 3 AI superpowers - Voice-First Tutor, 1-Click Kanban Auto-Scheduler, Multimodal Vision & GitHub Inspector
90a99d67	feat: Render rich Markdown in AI chat, auto-extract structured JSON, integrate transparent SVG/PNG logo